

Andre Aw

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EDUCATION

University of Toronto

Honors Bachelor of Science in Computer Science

Expected Graduation Jun 2027

Toronto, CA

EXPERIENCE

System Administrator Intern

Aug 2022 – Aug 2023

YSL Realty

Manila, PH

- Deployed and maintained local Linux servers for file storage and web hosting, reducing cloud hosting fees by 100%
- Set up a dashboard for secure file sharing with SFTP and SSH, reducing file sharing latency by 80%
- Eliminated accidental file overwrites using mtime-based versioning system
- Used Python to automate mass digitization of paper documents into searchable PDFs, reducing document search times by 50%
- Created Bash scripts to automate routine maintenance tasks on servers

IT Support Intern

May 2021 – Jun 2022

Five West Realty

Manila, PH

- Used Bash, Powershell, and Python scripts to automate common labor-intensive tasks, reducing time spent on routine tasks by 50%
- Ensured standardization of software across computers, reducing software versioning conflicts by 90%
- Configured Layer 2 network switches and local Linux servers
- Configured office computers for reliability, reducing problems from network conflicts by 90%
- Troubleshoot computer problems brought by staff, increasing office productivity by 10%

PROJECTS

Smart Air | *Android app for parents to track children's medication*

Oct 2025 – Nov 2025

- Built and led a group of 5 developers in implementing a fullstack Android app over a 5-week period
- Designed the frontend of the app using Kotlin for user input and adjustable settings
- Implemented a RESTful API in Java with Firebase as the data store
- Guided 4 developers with advanced Git workflows, reducing time spent on deployment by 50%
- Used Agile methodologies such as Scrum and Kanban to ensure 5% early delivery of features

Clausewitz Manifest | *Web app for searching info in complex strategy games*

Mar 2025 – Jul 2025

- Built a fullstack web app that displayed over 5 million datapoints from Paradox Interactive games
- Designed a frontend using HTML, CSS, and Javascript that used lazy loading to display data without reloading pages
- Programmed a production-grade backend in Rust that served API endpoints in sub-50ms time
- Sourced data from a Postgres database using the SQLX Rust library
- Hosted on AWS using EC2 instances with Docker containers for easy deployment and scalability

Clausewitz Parser | *Parser and database generator for domain-specific language*

Feb 2025 – Jun 2025

- Developed a production-grade parser in Rust for a domain-specific language used in Paradox Interactive games
- Achieved parsing speeds of 800MB/s per core and multithreaded computation using tokio
- Read parse output into a Postgres database using the SQLX Rust library
- Increased insert speed using SQL strategies such as dropping indices during insertion, using unlogged tables, and grouping batches of inserts into transactions, achieving 200MB/s insert speeds on 50GB text datasets

Nix Devenv | *Highly configured Linux systems using Nix*

Nov 2024 – Mar 2025

- Managed configuration of pseudo-containerized Linux/Unix environments
- Used a uniform infrastructure-as-code interface to configure bootloaders, systemd services, and program configuration
- Developed Bash and Python scripts for wrapping unpackaged binaries and automating workflows
- Used CI/CD to automatically build and deploy system configurations to multiple machines
- Reduced unrecoverable system breakages by 99% through atomic upgrades and both bootloader and Git-based rollbacks
- Ensured reproducibility by storing version info in lockfiles tracked by Git